TIME TRAVEL AND BECOMING

I wish to discuss a supposed implication of one sort of time travel. The sort of time travel is time travel into one's past along a closed timelike curve (CTC). The implication is that in spacetimes with CTCs there can be no temporal passage or "flow" of time. I will argue that the implication does not hold.

1. The Ontological (In)Significance of the Present

Kurt Gödel threw us a couple of curve balls. The first, his famous incompleteness proofs of first-order theories of arithmetic, decisively changed the course of the development of logic and foundations of mathematics.[1] The second, his discovery of an exact solution of the Einstein Field Equations representing a spacetime containing closed timelike curves is less well known.[2] A timelike curve in a general relativistic spacetime (or just spacetime, henceforth) represents a possible trajectory of a material body. Since the model spacetime that Gödel discovered is time orientable, it is possible to distinguish continuously at each point a future from a past direction in time. Since the curve is closed, like a circle, it is possible to begin a journey at a point on the curve, proceed along the curve and return to the original point. In addition, it is possible to make such a journey while traveling into the future at each point along the curve, so that one can depart from a point, travel into its future and yet return to it from its past. Each point in the spacetime that Gödel discovered lies on a CTC. At each point in the spacetime, therefore, it is possible to travel into its past by moving from it along an everywhere future-directed timelike curve.

At first there was only a trickle of comments on this result, mostly concerning the implications of this discovery for the possibility of time travel.[3] That trickle turned into a torrent when some physicists became seriously interested in time travel.[4] It was the great merit of Palle Yourgrau's 1991 book *The Disappearance of Time: Kurt Gödel and the Idealistic Tradition in Philosophy*[5] that it focused attention on the fact that Gödel (in his 1949b) drew a remarkable philosophical conclusion from the odd features of his newly discovered spacetime. The conclusion is that there is no such thing as the objective lapsing of time. Exactly how this conclusion is supposed to follow from his mathematical discovery is not altogether clear—a surprising claim to make given that Gödel is

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one of the greatest logicians ever.[6] But one premise in his argument is certainly the claim that in the model spacetime that he discovered there is "nothing corresponding to the passage of time". (39f) Here, finally, is the claim that I want to examine—the claim that in Gödel's model spacetime (or, more generally, in *any* spacetime with CTCs) there is "nothing corresponding to the passage of time".[7]

What's the argument for this claim? Here is what Yourgrau thinks that Gödel thinks:

> In a world with closed, circular time, then, time could not but be spacelike—that is, temporal position in it could not have the ontological significance that attends a genuine, open succession of stepwise generated times. *And that was, of course, exactly Gödel's point.* (p. 38)

The possibly mysterious phrase "ontological significance" is, I take it, at least partly explained earlier when Yourgrau says that "'Positions' in time are not 'ontologically neutral': the only location that confers the privilege of existence is that of the present, the now." (25) But there must be a bit more assumed for the notion of ontological significance in this argument. In open (non-circular) time, Yourgrau says, "when an earlier step has occurred, the later steps *have not* (they lie in the future, hence they are not yet actual or existent)." (38) So part of the ontological significance of a moment or step or stage of linear time is that it divides other moments into two disjoint classes, those that are not yet and those that are no longer.

A point on a CTC, it would seem, cannot do this. Given the usual (and natural) way that 'a is earlier than b' and 'b is later than a' are defined in relativistic spacetimes, these relations are both symmetric relations for events lying on a CTC. If e is some event on a CTC, then some other event e' on that CTC that is earlier than e has already happened as of e, yet (given the symmetry of this relation) e is also earlier than e', and so e' has not yet happened as of e. In particular, if the future is "open"—that is, if events to the future of or later than the present do not yet exist or are not definite—how could it be that these events are also past or prior to the present and so have already occurred? This situation seems impossible, and yet is also seems quite plausible that the passage of time in a spacetime with CTCs would require just this impossible situation to obtain. Since there are many situations in which CTCs do or may arise, then it seems we must follow Yourgrau (and Gödel, if this was indeed his reasoning) in concluding that in such spacetimes there can be no passage or "objective lapsing" of time.[8]

Yet there are three considerations that might give one pause before asserting this conclusion. The first consideration is that the argument relies in a relativistic context on a way of thinking about time appropriate to a pre-relativistic con-

text. Pre-relativistically (or classically) the relation '*a* is simultaneous with *b*' is an equivalence relation, and the resulting equivalence classes can be completely ordered by the relation '*a* is earlier than *b*', so that relative to some class of simultaneous events or to some representative element in that class, any other class of simultaneous events or the members of such a class are either earlier than it (the original chosen class or event) or later than it, but not both. However cumbersome or clumsy the formal expression of this idea is, it expresses a way of thinking about time that is deep in our habitual way of thinking about time, our "intuitions".

Nevertheless, this intuitive picture does not, as is well known, survive even the first step into relativistic physics, the special theory of relativity. In the spacetime of the special theory, the two-place relation '*a* is simultaneous with *b*' is not well defined, although the three-place relation '*a* is simultaneous with *b* relative to frame *F*' is. There is no unique or physically distinguished way to partition the events in special relativistic spacetime into equivalence classes of events, some earlier and some later than a given event or class of events. Rather, at each point of the spacetime, there is an uncountable infinity of distinct but physically equivalent ways to do this. This result is repugnant to the pre-relativistic intuitions evoked in the previous three paragraphs, yet it is a perfectly consistent picture of spacetime. The objection raised by Yourgrau merely points to a formal difference concerning some relations pre- and post-relativistically. Our pre-relativistic "intuitions" are not unbiased or authoritative arbiters of the nature of these relations.[9]

But, it will surely be objected, isn't this picture in special relativistic spacetime consistent precisely because it lacks the ingredient of real temporal passage and its "ontological significance"? How else could an event *b* be in one frame earlier and in another frame later than a given event *a*, unless the notions of earlier and later are drained of ontological significance? Similarly, even if Gödel's model spacetime is perfectly consistent, does it not too lack *real* time? Given a point *a* on a CTC, how can an event *b* on that CTC be both later and earlier than it? That is, yet again, how can something that doesn't yet exist (because it is later than a given event) already exist (since it is also earlier than that event)? The second consideration I would like to advance is that the notion of ontological significance evoked here, if it is itself significant, is quite difficult to state with any clarity or precision.

Consider Yourgrau's characterization of ontological significance that I quoted above. "'Positions' in time are not 'ontologically neutral': the only location that confers the privilege of existence is that of the present, the now." The only things that (really) exist, says Yourgrau, are those things that exist now. Suppose, however, that the verb 'exist' in this claim is a straightforward present tense verb. Then the sentence is (or comes to) a tautology. Only those things

that exist (now) exist (now). Suppose, though, that the first 'exist' in my paraphrase of Yourgrau's claim is what is often called the *tenseless* 'exist' (and which I call the *detensed* 'exist')—that it is really a disjunctive verb that should be read "has existed, exists now, or will exist". Then Yourgrau's characterization of ontological significance is no longer tautologous. It reads: "Only those things that exist now exist (tenselessly)". This claim is not easy to believe. Its truth would require, for instance, that either Isaac Newton never existed or that he exists now!

I hope that the reader who is not already familiar with the difficulties of stating significantly and persuasively what the "ontological significance" of the present comes to can now at least see that there is a *prima facie* problem in the enterprise. I think, in fact, that there is no characterization of this notion available along the lines of the one given above by Yourgrau, but this claim requires a far more detailed argument than I can develop here.[10] *If* the idea of "ontological significance" cannot be explicated along these lines, then the force of the claims about existence that were intended to motivate the discomfort with CTCs is moot. It is a trivial implication of the tensed/detensed ambiguity introduced in the last paragraph that one can consistently say that a given event *e* does not exist (tensed) but does exist (detensed). Adding 'already' or 'no longer' does not introduce any new inconsistency if we move from a global to a local perspective.

2. Passage Localized

To turn now to the third consideration relevant to evaluating Yourgrau's argument that there can be no passage or objective lapsing of time in a world with CTCs, we note that so far no effort has been made to characterize temporal passage. While this is a difficult notion indeed to capture, so that it is not entirely irresponsible to take it as primitive or basic, in this context the omission might seem surprising since Gödel himself presents a crisp characterization of it in the second paragraph of his 1949b paper. What Gödel says is:

> The existence of an objective lapse of time, however, means (or, at least, is equivalent to the fact) that reality consists of an infinity of layers of "now" which come into existence successively. (1949b, 558)

For starters, if we think in terms of Newtonian (or common-sense) spacetime, the layers of "now" are the successively occurring equivalence classes of simultaneous events discussed above. These layers are three-dimensional surfaces in a four-dimensional space (and are called *hypersurfaces*). Each layer represents all of space at an instant, and the entire spacetime is divided (partitioned, foliated) into a set of these layers. If, for ease of representation, we take these layers to be two-dimensional, we get a picture like this:

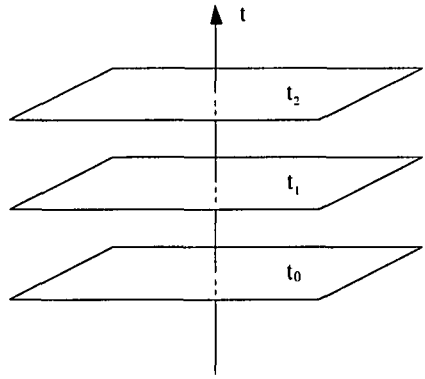


Figure 1: Successive sets of simultaneous events

It is sometimes supposed that this is a picture of a static world, a so-called "block universe", but this supposition is mistaken. The picture is static; but since the vertical dimension is time, what the picture represents is a universe evolving in time, the successive hyperplanes of simultaneity being the passage of time. One must not confuse features of the picture with features of the pictured. "We do not need an animated picture to have a picture of animation."[11]

It is worth emphasizing at this point that the hyperplanes of simultaneity in the diagram above are really intended to extend through all of space. They have to be given tidy boundaries in the illustration in order to fit on the page, but they may be infinite in extent. These layers of "now" are intended to represent sets of events occurring successively, the very march of time, throughout the universe. The picture is so natural that one might be startled by the thought that it represents not just a physical but also a large metaphysical commitment. My experience of the succession of events is entirely local. How do I, how could I, know what events in distant galaxies (or on the moon or even in Australia) are the ones happening *now*?

The epistemological critique of distant simultaneity was one of the routes to the special theory of relativity.[12] In the spacetime of the special theory, the unique classical foliation of the spacetime is replaced by an uncountable infinity of such foliations, no one of which can be distinguished from the others by any physical measurement or experiment. In the general theory, this picture is generalized even further. We can still foliate the spacetime with hyper-surfaces, but for our purposes what we should require of the "slices" is that (i) each point in a slice, Σ, is spacelike separated[13] from every other point in the slice and (ii) every inextendible timelike curve intersects the slice at exactly one point. Such slices are usually called *Cauchy surfaces*.

Cauchy surfaces, I believe, are what Gödel thought of as the formal counterparts in general relativistic spacetimes of his layers of "now". If one particular foliation could be singled out or distinguished from the infinity of other possible foliations, then one might well hold the distinguished foliation to represent the *objective* lapsing of time. Gödel noted in his 1949b that James Jeans had proposed a way that one particular foliation of spacetime might be singled out, based on the distribution of matter in the universe. Gödel's discovery of his new solution to the Einstein Field Equations took his argument with Jeans in a wholly unanticipated direction.

A moment's reflection will make it clear to the reader, if it isn't clear already, that in a part of spacetime in which there is a CTC there can be no Cauchy surfaces. If a CTC intersects a spacetime slice once, then it must do so again (in order for it to close). In Gödel's new model spacetime, every point is connected to itself by a CTC. Therefore, in Gödel's new model spacetime, there are no Cauchy surfaces at all. If the lapsing of time requires the existence of layers of "now" and these layers of "now" are (or are modeled by) Cauchy surfaces, then there is no lapsing of time or temporal becoming in Gödel's new model spacetime.[14] *This* connection between the presence of CTCs and the absence of passage, the reader will note, does not depend at all on the dodgy metaphysics of the ontological significance of the present.

The connection is not free of all metaphysics, however. Does the lapsing of time really require Cauchy surfaces, hyper-surfaces extended through all of space? Can this idea not be criticized in the same way that we criticized simultaneity slices in classical spacetimes? Our experience of time and the present is local. Cauchy surfaces are global. To suppose that Cauchy surfaces are necessary for passage is to outrun experience by a wide margin, however natural the supposition may feel.

Is there a way, then, to think of the passage of time as a succession of presents or nows, but locally? There may well be, if we can help ourselves to another tempting idea. The present is often represented as a surface with no thickness of extension, but the present for us is extended. Suppose, for the sake of simplicity, that the temporal extent of our psychological present is one second.[15] Suppose also that 'locally' means *along a timelike curve*, λ. Such curves can be parameterized in terms of a quantity known as *proper time*, τ. Consider some closed interval along λ of length 1 second (assuming that we choose the second as the unit of proper time), the endpoints of which we may label p and q, with p earlier than q.[16] During the second separating p and q (along λ), a light signal could leave p, travel 150,000 km, be reflected by a mirror, and then return to q. Consider then the set of events contained within the boundary defined by the paths of all possible such light signals leaving p and returning to q.[17] I would like to propose that this spatially wide but temporally squat set be *the present*

for event q. A series of events or points along a timelike curve λ would have a series of associated presents, as in the following picture. I suggest that this picture represents the passage of time along the timelike curve λ.

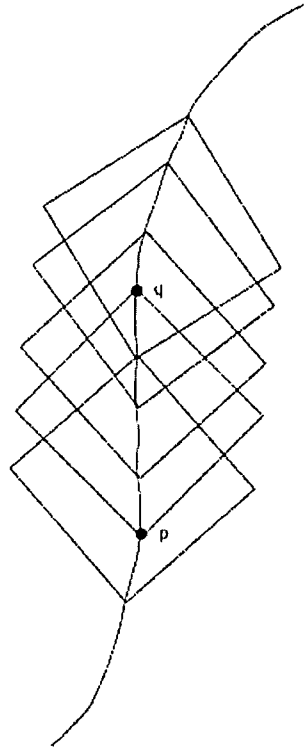


Figure 2: Successive presents along a timelike worldline

Is this picture coherent when the curve in question is a CTC? Now that we have dissociated ourselves from the suspect metaphysics of the universal now, what reason is there to suspect that it is not? All points on the curve equally exist in the detensed sense, which means only that they are points or events in the spacetime. Any point on the curve, were one located there, has associated with it a well-defined present. One might feel that only that point and its associated present exist (in the tensed sense). Others at other places and times moving on other timelike trajectories will disagree, but we are used to such disagreements in relativity theory. And for those who are nearby and moving slowly with respect to one another, the amount of overlap of their presents will be so

great that it will be easy to understand how we developed the idea of frame-dependent universe-wide layers of "now".[18]

If passage or temporal becoming can be understood in the minimalist or deflationary way I suggest, and if the present can be understood naturalistically and locally in the way depicted above, then even spacetimes that admit of time travel into the past along CTCs are not bereft of passage.[19]

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NOTES

1. Godel (1931).

2. The technical result appeared in Gödel (1949a). A philosophical discussion of the result appeared in Gödel (1949b).

3. See e.g. Malament (1984), Sklar (1984), and Horwich (1987), ch. 7. See also Stein (1970).

4. For a popular account of how this happened see Thorne (1994), ch. 14, "Wormholes and Time Machines"; for a philosophical account see Earman (1995), ch. 6, "Time Travel"; and for a recent review of attempts to show that CTCs are, after all, physically impossible, see Visser (2003).

5. The book was later re-issued with some additions as Yourgrau (1999). Page references in the text will be to the later edition, unless otherwise noted.

6. But the point of the Appendix to ch. 6 in Earman (1995).

7. If one thinks that there can be time only if there is passage, then, as Yourgrau (1999: 71) points out, one is in the odd situation of arguing that the possibility of time travel along CTCs implies that there is no time!

8. See Visser (2003) above for a short catalog of other sorts of spacetimes with CTCs. He points out, by the way, that "for all practical purposes the phrases 'time machine' and 'closed timelike curve' . . . can be used interchangeably." (p. 161).

9. Of course, when event *a* is connected to event *b* by a smooth, everywhere future-directed timelike curve, the relation '*a* is earlier than *b*' is frame-independent. This observation is no objection to my argument, if the second and third considerations below are on the right track.

10. I develop this argument in my (forthcoming).

11. As I wrote in my (2002: 163).

12. See Sklar (1974), ch. IV, for an account of the details.

13. Two points in a spacetime are spacelike separated if no light signal or material object can travel from one to the other. Given this constraint, our general relativistic spacetime-wide slices no longer need look flat, like the picture from classical spacetime. They may be hilly or curvy, as long as the hills are not too steep, the curves not to sharp. This intuitive description can be made precise of course.

14. I developed this point in more detail in my own paper assessing Gödel's argument (Savitt 1994).

15. I recently heard an eminent cognitive psychologist say that the psychologi-

cal present (aka the specious present) is typically given bounds of .5 seconds to 3 seconds.

16. This remark assumes an oriented spacetime. Gödel's model spacetime is orientable, and we will assume here a given orientation.

17. Formally, the set defined is usually indicated by $I^+(p) \cap I^-(q)$. This set is the intersection of the interior of the future light cone of p with the interior of the past light cone of q. In the language of Naber (1988: 29), this set is the intersection of the future time cone of p with the past time cone of q. The existence of this set of events is not dependent upon any sort of mind or sensibility "experiencing" the events p and q, although the length of proper time chosen to separate the beginning and end of each present is fixed with an eye towards human psychology. It might be better to think simply of a set of (extended) presents along a timelike curve than a set of presents attached to or attributed to points or events.

18. Especially if one attends to Stein (1991). I recently learned that Richard Arthur has come to similar conclusions through a similar route, and still more recently that Dennis Dieks, Mauro Dorato, and Simon Saunders also have developed similar ideas concerning time in relativity theory.

19. I am grateful to Kristin Schleich of the UBC physics department for the patient and helpful way she answered my many questions about the general theory of relativity, and the Social Sciences and Humanities Research Council of Canada for their support of my research.

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